

Task 9 — Experimentation Platform, Feature Flags & Guardrails

Data Analyst · Sprint B - Growth & Experimentation · Task Brief

DATA ANALYST

PlaceMux · Phase 3 · Task Brief

TASK 9 / 25

Deep Specialist · Expert

Sprint B - Growth & Experimentation · Mastery 73%

Focus

Build the experimentation analysis framework: power, significance, SRM checks and a decision protocol.

What to build / complete

- An experiment analysis framework (power calc, significance, CIs, SRM check)
- A pre-registration and decision protocol the team must follow
- Automated experiment readouts

Execution — work through in order

Phase 3 gives you a goal and a bar, not a recipe. These stages are the discipline; the design decisions are yours to make and defend.

Stage A — Understand & set the bar

1. Re-read the Expected Output and the scoring parameters so you build to the bar, not past it.
2. Confirm every prerequisite and upstream input is actually in your hands, not merely promised.
3. Restate, in one sentence, what 'good' means here: The company can no longer fool itself with a bad experiment, because the framework will not let it.

Stage B — Build: An experiment analysis framework (power calc, significance, CIs, SRM check)

1. Define the metric, its source & the decision

Define precisely what “An experiment analysis framework (power calc, significance, CIs, SRM check)” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage C — Build: A pre-registration and decision protocol the team must follow

1. Define the metric, its source & the decision

Define precisely what “A pre-registration and decision protocol the team must follow” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage D — Build: Automated experiment readouts

1. Define the metric, its source & the decision

Define precisely what “Automated experiment readouts” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage E — Integrate, verify & make demoable

1. Wire your work into the end-to-end flow and run the whole journey once, start to finish.
2. Run a real experiment through the framework, including a deliberate SRM detection.
3. Break it on purpose: force the failure path and confirm it degrades the way you designed.
4. Prepare a live demo on real data. Not slides, not screenshots, not 'it works on my machine'.

Done when (demoable by end of task)

- “An experiment analysis framework (power calc, significance, CIs, SRM check)” is complete, real, and demoable end-to-end.
- “A pre-registration and decision protocol the team must follow” is complete, real, and demoable end-to-end.
- “Automated experiment readouts” is complete, real, and demoable end-to-end.
- Good looks like: The company can no longer fool itself with a bad experiment, because the framework will not let it.

Worry if any of these are true

- Experiments run without a power calculation.
- No SRM check.
- Calling an experiment early because it looked good on day two.

Hand-off

- Experiment protocol to every role. Make sure the next person can pick it up without you.